Patch Model of Grass distribution in Salt Marshes

Fatima Islam Olivier Grimard

December 2024

1 Abstract

This report introduces a mathematical model utilizing first-order nonlinear Ordinary Differential Equations (ODEs) to describe the distribution and interaction of two grass species, Spartina alterniflora and Spartina patens, across a salt marsh. The model simplifies the marsh's stress gradient into discrete patches and accounts for interspecific and intraspecific competition, as well as species-specific resistance to stress. Analytical proofs establish the existence of an unstable trivial equilibrium and a non-trivial equilibrium under specific and general parameter conditions, respectively. Numerical simulations confirm the existence of a non-trivial solution and investigate the effects of key parameters and initial state variables on this solution. While the model successfully replicates some expected patterns, we were unable to demonstrate a globally stable and ecologically coherent equilibrium. This analysis suggests potential changes to how competition and movement between patches are modeled.

2 Introduction

In ecology, the natural environment is often represented as discrete patches, particularly for heterogeneous environments, where parts of the environment are more suitable than the rest. Such is the case for the Theory of Island Biogeography developed by Robert MacArthur and Edward O. Wilson, where islands are considered suitable patch habitats in a matrix of unsuitable sea habitat [MacArthur and Wilson, 2001]. Isolation of these patches from one another restricts the movement of individuals between them and leads to each patch having distinct population dynamics. These populations can be considered separately and in unison as a metapopulation, a concept brought forward by Levin in metapopulation ecology [Levins, 1969]. The concept of patches has long been used to describe less obviously patchy environments. For instance, the role of patches in succession and disturbance [Pickett and White, 1985]. Any habitat disturbance is generally limited in scope within that habitat, forming patches of disturbed habitat within a matrix of undisturbed habitat. Sequential disturbance and succession processes lead these patches to form a speckled landscape of varying habitats. For example, forest fires create large gaps in the forest canopy which are then filled by forbs, shrubs, and trees in that order, with fires occurring at different times throughout the forest landscape; patches at varying stages of succession paint the forest matrix, creating heterogeneity within the ecosystem [Turner, 1989]. That being said, the use of patches is not restricted to heterogeneous environments where patches can be well defined.

Patches can also be used as a discrete representation of an ecosystem with gradients of abiotic constraints or resources. In large ecosystems that vary over a gradient, it is often impractical to obtain measurements throughout the gradient. Instead, measurements are collected at intervals along this gradient, simplifying the system into "patches" [Wiens, 1976]. When modeling populations living within a gradient, setting both time and space as continuous forces the use of more complex mathematical tools (i.e., PDEs) which might not yield much better outcomes for the trouble. As such, similar to field practices, models often involve simplifying the approach to multiple patches. Theoretically, it is often useful to present a habitat with a set of characteristics with fixed values such that its state is more easily understood than when having to consider the way these values change within the environment. This theoretical simplification is often used for salt marshes.

Salt marshes are soft sediment intertidal ecosystems dominated by salt-tolerant plants [Bertness and Ellison, 1987]. Although generally possessing a complex topography, salt marshes are often thought of as downward sloping from their landward side to their seaward side. The lower part of the salt marsh near the sea is more often inundated by the tides whilst the landward side is less so. Inundation is the dominant form of stress for plants in a salt marsh due to depleted oxygen, accumulation of salt and sulfide, a harmful compound, in waterlogged soil [Irving A. Mendelssohn and James T. Morris, 2000]. A gradient of stress is thus formed, strongest at the seaward edge and diminishing towards the landward edge. But instead of a gradient, salt marshes are often described in two discrete zones, high and low marsh. These two zones are dominated by two different types of grasses, a result of the different adaptations of the grasses residing in salt marshes. Along the North American Atlantic coast *Spartina alterniflora* is found in the low marsh, sections of the marsh more frequently inundated. *S. alterniflora* is well adapted to these flooded conditions but actually grows best in the favorable conditions of the high marsh. The high marsh is dominated by *Spartina patens*, which generally outcompetes *S. alterniflora* in this location but cannot grow in the low marsh as it cannot survive the stress of frequent inundations [Mark D. Bertness and Steven C. Pennings, 2000]. As such, when both species are present, *S. alterniflora* is restricted to the low marsh and is found sparsely through the high marsh, while *S. patens* grows only in the high marsh [Bertness, 1991].

In this report, we introduce a first-order nonlinear Ordinary Differential Equations (ODE) model to describe the distribution of these two plant species across a discretized gradient. An ODE involves one independent variable (generally time) and one dependent variable where an equation describes the derivative of the dependent variable with respect to the independent variable (i.e., the rate of change of the dependent variable with respect to the independent variable). It is of first order as it involves only first-order derivatives and is nonlinear since some dependent variables have powers other than one. With only two patches, the model reflects the low and high marsh but can scale to more patches for finer gradient detail. Each ODE in this system represents the growth rate of a species in a specific patch, forming a network of equations that collectively describe the dynamics of these two species across the marsh. The ODE incorporates intra- and interspecific competition, as well as the effect of stress on plant mortality, which varies depending on each species' resistance to that stress. We hypothesize that this model will accurately predict the distribution of *S. alterniflora* and *S. patens* in its simplest two-patch form and maintain accuracy with increased patch resolution, given the right parameter values are found . We want to prove two things of ecological relevance first seeing the parttern as describe above and that equilibrium being a globally stable equilibrium or at least reacheable from an initial dominance of *S. alterniflora*

3 Model

3.1 General Model

$$\frac{dA_i}{dt} = \sum_{n=1}^{\rho} \frac{g_a A_n}{|n-i|+1} \left(\frac{K_a}{K_a + A_i + P_i} \right) - \frac{d_a A_i \phi_i}{R_a K_a}, \quad i = 1, 2, \ldots, \rho$$
$$\frac{dP_i}{dt} = \sum_{n=1}^{\rho} \frac{g_a P_n}{|n-i|+1} \left(\frac{K_p}{K_p + P_i + A_i} \right) - \frac{d_p P_i \phi_i}{R_p K_p}, \quad i = 1, 2, \ldots, \rho$$

where:

- ρ is the number of patches considered.
- A_i and P_i are the population of *Spartina Alterniflora* and *S. patens* in patch i, respectively.
- g_a and g_p are the per capita growth rate of *S.Alterniflora* and *S.patens*, respectively.
- K_a and K_p represent the half-saturation constants or Michaelis-Menten constants their respective species.
- d_a and d_p are the per capita mortality rate of both species respectively.
- ϕ_i indicates the abiotic stress level of the i^{th} patch.
- R_a and R_p are unique level of resistance to abiotic stress of each species.

3.2 Two-patch Model

For analysis,we consider the 2-patch model as in the form,

$$\frac{dA_1}{dt} = \left(g_a A_1 + \frac{g_a A_2}{2} \right) \left(\frac{K_a}{K_a + A_1 + P_1} \right) - \frac{d_a A_1 \phi_1}{R_a K_a}$$
$$\frac{dP_1}{dt} = \left(g_p P_1 + \frac{g_p P_2}{2} \right) \left(\frac{K_p}{K_p + P_1 + A_1} \right) - \frac{d_p P_1 \phi_1}{R_p K_p}$$
$$\frac{dA_2}{dt} = \left(g_a A_2 + \frac{g_a A_1}{2} \right) \left(\frac{K_a}{K_a + A_2 + P_2} \right) - \frac{d_a A_2 \phi_2}{r_a K_a}$$
$$\frac{dP_2}{dt} = \left(g_p P_2 + \frac{g_p P_1}{2} \right) \left(\frac{K_p}{K_p + P_2 + A_2} \right) - \frac{d_p P_2 \phi_2}{r_p K_p}.$$

Theorem 3.1. *The system dynamics are governed by the fixed-point equation:*

$$x = FD^{-1}x, \tag{3.1}$$

where:

- $x = \begin{bmatrix} A_1 \\ P_1 \\ A_2 \\ P_2 \end{bmatrix}$ *is the state vector.*
- $F(A)$ *is the birth matrix derived from interspecies growth interactions.*

- *$D(A)$ is the death matrix representing species-specific mortality.*

It has

1. *a unique fixed point x^*, which represents the equilibrium of the system,is locally and globally stable.*
2. *unstable equilibrium point,x_0.*

Proof. The matrix from the model is defined as[[Caswell, 2001]],

$$F = \begin{bmatrix} f_{11} & f_{12} & 0 & 0 \\ f_{21} & f_{22} & 0 & 0 \\ 0 & 0 & f_{33} & f_{34} \\ 0 & 0 & f_{43} & f_{44} \end{bmatrix},$$

where:

$$f_{11} = \frac{g_a K_a}{K_a + A_1 + P_1}, \quad f_{12} = \frac{g_a K_a}{2(K_a + A_2 + P_1)}, \quad f_{21} = \frac{g_a K_a}{2(K_a + A_1 + P_2)}, \quad f_{22} = \frac{g_a K_a}{K_a + A_2 + P_2},$$
$$f_{33} = \frac{g_p K_p}{K_p + P_1 + A_1}, \quad f_{34} = \frac{g_p K_p}{2(K_p + P_2 + A_1)}, \quad f_{43} = \frac{g_p K_p}{2(K_p + P_1 + A_2)}, \quad f_{44} = \frac{g_p K_p}{K_p + P_2 + A_2}.$$

The death matrix D is diagonal and given by:

$$D = \begin{bmatrix} \frac{d_a \phi_1}{R_a K_a} & 0 & 0 & 0 \\ 0 & \frac{d_a \phi_1}{R_a K_a} & 0 & 0 \\ 0 & 0 & \frac{d_p \phi_2}{R_p K_p} & 0 \\ 0 & 0 & 0 & \frac{d_p \phi_2}{R_p K_p} \end{bmatrix},$$

with its inverse:

$$D^{-1} = \begin{bmatrix} \frac{K_a R_a}{d_a \phi_1} & 0 & 0 & 0 \\ 0 & \frac{K_a R_a}{d_a \phi_1} & 0 & 0 \\ 0 & 0 & \frac{K_p R_p}{d_p \phi_2} & 0 \\ 0 & 0 & 0 & \frac{K_p R_p}{d_p \phi_2} \end{bmatrix}.$$

So,

$$G = FD^{-1} = \begin{bmatrix} \frac{K_a R_a}{d_a \phi_1} \frac{g_a K_a}{K_a + A_1 + P_1} & \frac{K_a R_a}{d_a \phi_1} \frac{g_a K_a}{2(K_a + A_2 + P_1)} & 0 & 0 \\ \frac{K_a R_a}{d_a \phi_1} \frac{g_a K_a}{2(K_a + A_1 + P_2)} & \frac{K_a R_a}{d_a \phi_1} \frac{g_a K_a}{K_a + A_2 + P_2} & 0 & 0 \\ 0 & 0 & \frac{K_p R_p}{d_p \phi_2} \frac{g_p K_p}{K_p + P_1 + A_1} & \frac{K_p R_p}{d_p \phi_2} \frac{g_p K_p}{2(K_p + P_2 + A_1)} \\ 0 & 0 & \frac{K_p R_p}{d_p \phi_2} \frac{g_p K_p}{2(K_p + P_1 + A_2)} & \frac{K_p R_p}{d_p \phi_2} \frac{g_p K_p}{K_p + P_2 + A_2} \end{bmatrix}.$$

Each term in the matrix can be interpreted as follows[3],

- The diagonal terms (G_{11}, G_{22},G_{33} and G_{44}) represent self-regulation, i.e., intra-species interactions.
- The off-diagonal terms (G_{12} , G_{21},G_{34} and G_{43}) capture inter-species effects.

delete the subtracted diagonal terms

The objective is to prove that the equilibrium point of this system is unique and locally asymptotically stable.This is achieved through spectral radius analysis, Gershgorin's Circle Theorem, diagonal dominance, and the Perron-Frobenius theorem.

The uniqueness of the equilibrium is proven in three stages: **Spectral radius analysis[[Caswell, 2001], [Horn and Johnson, 2012]],Perron-Frobenius theorem application**[[**Meyer, 2000**]], and **Contraction mapping[[Hirsch et al., 2013]]**.Uniqueness of the equilibrium is established by showing that the spectral radius $\rho(G) < 1$. This ensures that the system exhibits a contraction mapping, implying a unique fixed point.

3.3 Spectral Radius Analysis

The spectral radius $\rho(G)$ is defined as the largest magnitude of the eigenvalues of $F(A)D(A)^{-1}$:

$$\rho(G) = \max\{|\lambda| : \lambda \text{ is an eigenvalue of } G\}. \tag{3.2}$$

3.3.1 Characteristic Equation

The eigenvalues λ satisfy the characteristic equation:

$$\det(G - \lambda I) = 0, \tag{3.3}$$

where I is the identity matrix. The determinant can be calculated by considering the block-diagonal structure of G. Each block determinant expands as

$$\det \begin{bmatrix} G_{11} - \lambda & a_{12} \\ G_{21} & G_{22} - \lambda \end{bmatrix} = (G_{11} - \lambda)(G_{22} - \lambda) - G_{12}G_{21},$$
$$\det \begin{bmatrix} G_{33} - \lambda & G_{34} \\ G_{43} & G_{44} - \lambda \end{bmatrix} = (G_{33} - \lambda)(G_{44} - \lambda) - G_{34}G_{43}.$$

Combining these, the characteristic equation for $F(A)D(A)^{-1}$ becomes:

$$\big[(G_{11} - \lambda)(G_{22} - \lambda) - G_{12}G_{21}\big] \cdot \big[(G_{33} - \lambda)(G_{44} - \lambda) - G_{34}G_{43}\big] = 0. \tag{3.4}$$

If all eigenvalues have magnitudes less than, the perturbations in x,state vector decay over time, leading to stability.

FD^{-1} is a function of x. What do the eigenvalues (which will also be functions of x) tell us about solutions?

3.3.2 Eigenvalue Bounds

Using the trace and determinant relations:

$$\lambda_1 + \lambda_2 = \mathrm{tr}(G) = G_{11} + G_{22} + G_{33} + G_{44}, \tag{3.5}$$
$$\lambda_1\lambda_2 = \det(G) = (G_{11}G_{22} - G_{12}G_{21})(G_{33}G_{44} - G_{34}G_{43}). \tag{3.6}$$

Using Gershgorin's Circle Theorem, we can bound the eigenvalues of G. Each eigenvalue λ_i lies within a disk centered at the diagonal entry G_{ii} with radius equal to the sum of the absolute values of the off-diagonal entries in the corresponding row.

$$|\lambda - G_{ii}| \leq \sum_{j \neq i} |G_{ij}|. \tag{3.7}$$

The radius of the disk is:

$$\sum_{j \neq 1} |G_{1j}| = |G_{12}|.$$

Since $G_{11} > G_{12}$ for all feasible values of A_i, P_i, the eigenvalues are bounded by the diagonal dominance of G.

Given the structure of $F(A)D(A)^{-1}$, we can show that:

$$\rho(FD^{-1}) \leq \max\left(G_{ii} + \sum_{j \neq i} |G_{ij}|\right)$$

3.3.3 Proof of Diagonal Dominance

To prove diagonal dominance [[Meyer, 2000]], for the matrix FD^{-1}, we verify that the magnitude of each diagonal element is greater than the sum of the magnitudes of the off-diagonal elements in the corresponding row.

$$G_{ii} > \sum_{j \neq i} G_{ij}, \quad \text{for } i \in \{1, 2, 3, 4\}.$$

For instance, in row 1,

$$G_{11} = \frac{K_a R_a}{d_a \phi_1} \frac{g_a K_a}{K_a + A_1 + P_1},$$
$$G_{12} = \frac{K_a R_a}{d_a \phi_1} \frac{g_a K_a}{2(K_a + A_2 + P_1)}.$$

Since $2(K_a + A_2 + P_1) > (K_a + A_1 + P_1)$, it follows that $G_{11} > G_{12}$.

General Formulation,

- G_{ij} terms are proportional to $\frac{g_a K_a}{K_a + A + P}$, which decay as $A, P \to \infty$.

Repeating this argument for all rows shows that FD^{-1} is diagonally dominant. Implication for Stability. Since G is diagonally dominant, the spectral radius satisfies:

$$\rho(G) < 1,$$

ensuring stability of the equilibrium point.

3.4 Perron-Frobenius Theorem

The matrix G is non-negative because:

- F represents birth rates, which are non-negative by definition.
- D^{-1} is the inverse of the positive diagonal matrix D, and hence its entries are positive.

By the Perron-Frobenius theorem:

- A non-negative matrix such as G has a unique largest eigenvalue $\lambda_{\max} > 0$.
- The eigenvector corresponding to $\lambda_{\max}$ is strictly positive, ensuring a biologically feasible equilibrium.

The equilibrium condition is defined as:

$$(F - D)x = 0.$$

This equation implies that equilibrium populations are reached when the net growth, represented by $F - D$, becomes zero.

Connection to the Perron-Frobenius Theorem The relationship between the principle eigenvalue $\lambda_{\max}$ of FD^{-1} and equilibrium can be established as follows:

- If the matrix FD^{-1} has $\lambda_{\max} = 1$, the population dynamics balance births and deaths at equilibrium.
- The equilibrium solution, x^*, satisfies:
$$(F - D)x^* = 0,$$
where x^* is a right eigenvector of $F(x^*) - D$ corresponding to $\lambda = 1$.

Diagonal Dominance and Stability Diagonal dominance and monotonic decay of off-diagonal terms (G_{12}, G_{21}) reinforce uniqueness:

- Diagonal Dominance: If the diagonal terms $(G_{11}, G_{22}, G_{33}, G_{44})$ dominate the sum of the off-diagonal terms $(G_{12}, G_{21}, G_{34}, G_{43})$, the spectral radius $\rho(FD^{-1})$ satisfies,

$$\rho(FD^{-1}) < 1,$$

  ensuring x^* is locally asymptotically stable.

- Monotonic Decay: The off-diagonal terms $G_{12}G_{12}, G_{21}, G_{34}$ and G_{43}, which represent interspecies interactions, decay as A and P increase, reducing their influence in the matrix.

Thus, the Perron-Frobenius theorem guarantees that:

- The largest eigenvalue $\lambda_{\max} = 1$ corresponds to the unique equilibrium.
- No other eigenvalue equals 1, ensuring the equilibrium is unique and biologically feasible.

3.5 Contraction Mapping

The fixed-point equation:

$$x = FD^{-1}x, \tag{3.8}$$

defines a mapping:

$$T(x) = FD^{-1}x. \tag{3.9}$$

To prove that $T(x)$ is a contraction[6], we examine the difference:

$$\|T(x_1) - T(x_2)\| = \|FD^{-1}(x_1 - x_2)\|.$$

Using the submultiplicative property of norms:

$$\|T(x_1) - T(x_2)\| \leq \|FD^{-1}\|\|x_1 - x_2\|.$$

The spectral radius of FD^{-1}, denoted $\rho(FD^{-1})$, satisfies:

$$\|FD^{-1}\| \geq \rho(FD^{-1}).$$

Since $\rho(FD^{-1}) < 1$, the mapping $T(x)$ is a contraction for all $x_1, x_2 \in \mathcal{D}$, where $\mathcal{D}$ is the biologically feasible domain of x (non-negative populations). The entries of F and D are bounded due to the form:

$$f_{ij} = \frac{gK}{K + A + P},$$

which decays as $A, P \to \infty$. The entries of D^{-1} are positive constants:

$$D_{ii}^{-1} = \frac{KR}{d\phi}.$$

Thus, the norm $\|FD^{-1}\|$ is uniformly bounded.

By the Banach Fixed-Point Theorem[[Deimling, 1985, Khalil, 2002]], $T(x)$ has a unique fixed point x^*, and all trajectories starting in $\mathcal{D}$ converge to x^*.

The spectral radius condition $\rho(FD^{-1}) < 1$ ensures that perturbations from the equilibrium decay over time, leading to convergence to x^*. The structure of FD^{-1}, derived from birth and death rates, ensures $\rho(FD^{-1}) < 1$ under biologically realistic conditions,

- Diagonal dominance of FD^{-1} ensures self-regulation dominates interspecies interactions.
- Monotonic decay of off-diagonal terms with increasing populations reduces interspecies effects, further reinforcing stability.

The uniqueness of x^* and the contraction mapping guarantee that:

$$x(t) \to x^* \quad \text{as } t \to \infty,$$

for any initial condition $x(0) \in \mathcal{D}$.

So,by spectral radius analysis, Perron-Frobenius theorem, and contraction mapping principles, the equilibrium point x^* is proven to be unique and Globally aymptotically stable. The structure of FD^{-1} ensures that the system dynamics converge to this equilibrium.

i.e x^* is a unique locally and globally asymptotically stable fixed point. □

Proof. For instability, there exists at least one eigenvalue $\lambda > 0$ [8]when,

- The spectral radius $\rho(FD^{-1}) > 1$.
- The off-diagonal terms (e.g., interspecies interactions a_{ij}) dominate diagonal self-regulation terms a_{ii}.

Now, At $x = 0$,matrix G has the following structure,

$$G_0 = \begin{bmatrix} \frac{K_aR_ag_a}{d_a\phi_1} & \frac{K_aR_ag_a}{2d_a\phi_1} & 0 & 0 \\ \frac{K_aR_ag_a}{2d_a\phi_1} & \frac{K_aR_ag_a}{d_a\phi_1} & 0 & 0 \\ 0 & 0 & \frac{K_pR_pg_p}{d_p\phi_2} & \frac{K_pR_pg_p}{2d_p\phi_2} \\ 0 & 0 & \frac{K_pR_pg_p}{2d_p\phi_2} & \frac{K_pR_pg_p}{d_p\phi_2} \end{bmatrix}.$$

This matrix describes the interaction between birth and mortality rates at $x = 0$. It has a block-diagonal structure:

$$G_0 = \begin{bmatrix} M & 0 \\ 0 & N \end{bmatrix},$$

where:

$$M = \begin{bmatrix} \frac{g_aK_aR_a}{d_a\phi_1} & \frac{g_aK_aR_a}{2d_a\phi_1} \\ \frac{g_aK_aR_a}{2d_a\phi_1} & \frac{g_aK_aR_a}{d_a\phi_1} \end{bmatrix}, \quad N = \begin{bmatrix} \frac{g_pK_pR_p}{d_p\phi_2} & \frac{g_pK_pR_p}{2d_p\phi_2} \\ \frac{g_pK_pR_p}{2d_p\phi_2} & \frac{g_pK_pR_p}{d_p\phi_2} \end{bmatrix}.$$

Now, Eigenvalues For M and N matrix,

The eigenvalues of G_0 are the union of the eigenvalues of M and N, as FD^{-1} is block diagonal:

$$\text{Eigenvalues of } M : \lambda_1, \lambda_2, \quad \text{Eigenvalues of } N : \lambda_3, \lambda_4.$$

The characteristic equation of M is:

$$\det(M - \lambda I) = \left(\frac{g_aK_aR_a}{d_a\phi_1} - \lambda\right)^2 - \left(\frac{g_aK_aR_a}{2d_a\phi_1}\right)^2 = 0.$$

Simplify:

$$\lambda^2 - 2\lambda\frac{g_aK_aR_a}{d_a\phi_1} + \frac{3}{4}\left(\frac{g_aK_aR_a}{d_a\phi_1}\right)^2 = 0.$$

The solutions are:

$$\lambda_1 = \frac{3g_aK_aR_a}{2d_a\phi_1}, \quad \lambda_2 = \frac{g_aK_aR_a}{2d_a\phi_1}.$$

Similarly, for N:

$$\lambda_3 = \frac{3g_pK_pR_p}{2d_p\phi_2}, \quad \lambda_4 = \frac{g_pK_pR_p}{2d_p\phi_2}.$$

The spectral radius of G_0 is the largest eigenvalue:

$$\rho(G_0) = \max(\lambda_1, \lambda_2, \lambda_3, \lambda_4).$$

Now, The Jacobian matrix at $x = 0$ is:

$$J = G_0 - I.$$

The eigenvalues of J are:

$$\mu_i = \lambda_i - 1, \quad \text{for } i = 1, 2, 3, 4.$$

If $\lambda_i > 1$, then $\mu_i > 0$, indicating instability and that will be when,

1. If reproduction dominates mortality ($g_a > d_a$ or $g_p > d_p$), the spectral radius $\rho(FD^{-1}) > 1$.
2. This implies that small perturbations away from $x = 0$ grow exponentially, making $x = 0$ an unstable equilibrium.

So,the equilibrium $x = 0$ is unstable if:

$$\rho(G_0) = \max\left(\frac{3g_aK_aR_a}{2d_a\phi_1}, \frac{3g_pK_pR_p}{2d_p\phi_2}\right) > 1.$$

So,Unstable non-trivial equilibria exist if

$$\lambda_{\max}(FD^{-1}) > 1.$$

Trivial Equilibrium ($x = 0$): The trivial equilibrium $x = 0$ is always unstable if $g_i > d_i$, meaning reproduction dominates mortality. Biologically, extinction states are unstable as populations grow from initial positive densities. □

Mathematical textbook used as reference for this part of the report are [Horn and Johnson, 2012, Hirsch et al., 2013, Murray, 2002, Meyer, 2000, Caswell, 2001, Kreyszig, 1989, Deimling, 1985, Khalil, 2002]

4 Numerical Analysis

Numerical analysis was conducted in R [R Core Team, 2023] with packages deSolve [Karline Soetaert et al., 2010] and rootSolve [Karline Soetaert, 2009]. The code for the numerical analysis is located in the GitHub repository. It consists of 4 R files: "functions.r" in which all functions used are outlined, "model.r" in which the function describing the model are found, "computation.r" where the functions are executed, and "extracode.r" which has code for nullcline computation, initial code development for vector fields, and others. The code in extracode was not pursued further following Dr. Watmough's suggestion of avoiding nullcline and vector field in understanding high-dimensional models. The code should be run on a Unix-based system given the method chosen for parallelization of costly computation, using the parallel package [R Core Team, 2024] in R. Most of the numerical analysis has been developed to work with any number of patches, but given the limited timeframe for the realization of this report, we will present only the two-patch model. Much more analysis was done than is presented, since interpretation of results was often confusing or the analysis was not successful in finding the desired results. Hence, the report will focus on solution dynamics in relation to $R_{a,p}$, ϕ_i, and the initial population conditions. The ecological coherence of solutions is briefly discussed, and improvement of the model is suggested. Ecological coherence discussed was assumed to require parameters with the following set of rules:

$$g_p > g_a, \frac{d_p}{R_p} > \frac{d_a}{R_a}, K_p > K_a, d_{a,p} \neq 0, g_{a,p} \neq 0$$

and an equilibrium where:

$$P_2 > A_2\ ,\ A_2 < A_1\ ,\ A_1 > P_1$$

4.1 Solution under varying parameters

Given that each species had unique death and growth rates, the parameters worth looking at in this analysis were thought to be species resistance and the patch stress ϕ. Patch stress level is an important parameter, being the only one differentiating one patch from the other. Consider that the solutions were generated with the following base parameter values in Table 2, differing only in the parameter under inspection in the figures. The general initial state variables used when varying R and ϕ were chosen for their simplicity, but alternative starting points will be inspected. Note that growth and death rates in this section do not mean

$g_{a,p}$ and $d_{a,p}$ but rather the positive and negative elements of the ODE function, respectively. Note also that in the code ϕ is called "h".

A general, ecologically coherent (although not perfectly for reasons outlined below) solution is outlined in Figure 1. Note that if the duration of the solution were to be elongated, the solution would not stay at an equilibrium, since the parameters are not precisely the right ones. R_p changes the dynamics of the system predictably, where higher R_p leads to the dominance of *S. patens* in both patches, which means that its growth was restricted sufficiently in patches 1 and 2 for *S. alterniflora* to outcompete it, but given reduced overall death rates, *S. patens* outcompetes *S. alterniflora*. The effect of this increase seems to affect P_2 more than P_1, which seems to indicate that the effect of increased resistance is greater when under less stress. I believe this points to a non-linear relationship between growth and resistance. As R_p reduces, the death rate of $P_{1,2}$ becomes higher than its growth rate, leading to its disappearance from both patches, although this is probably exacerbated by competition with $A_{1,2}$ (Figure 2). Similarly, a decrease in R_a leads *S. alterniflora* to disappear from both patches (Figure 3), likely as a combination of excess death rate due to reduced resistance to stress and competition with *S. patens*, limiting the growth rate. Interestingly, the change in ϕ seems to transform our coexistence equilibrium into a two-way equilibrium where one species dominates over the other, but changes in the most stressful patch (patch 1) seem to affect the dynamics less than changes in the least stressful patch (patch 2) (Figure 4). Increasing the stress of the second patch leads to the disappearance of *S. patens* from it and the establishment of *S. alterniflora*. Increasing its ϕ further leads to a decrease in *S. alterniflora* at equilibrium, which is consistent with the simple logic that increased death rates lead to lower population equilibrium points. Figure 4 also illustrates well the difficulty of finding the parameters that lead to coexistence, where most parameters will reach equilibrium where one species dominates both patches, fewer will have three-patch equilibrium, and even fewer will have full coexistence. This is also consistent with other tests made, although whether they truly tested this is debatable.

The last important point to consider is the effect of different initial conditions. Four different equilibria can be reached by varying the initial state variable values (Figure 5). Column 4 shows all the same equilibrium, with this equilibrium being reached if *S. patens* started at least as high as *S. alterniflora*, which is the case in this entire column. The same is true for column 2, where patch 1 *S. alterniflora* starts higher in all graphs but still falls low as a result of high abiotic stress in patch 1. For both of these rows, I am thinking the reason why *S. patens* is outcompeting *S. alterniflora* in patch 1 is due to its increased growth rate as a result of the large population in patch 2, which would always get much higher than *S. alterniflora* given the starting condition of $A_2 =$ or $< P_2$. This can be emphasized further by comparing the first two rows of column 3 in Figure 5, where the only difference is that patch 2 started with fewer *S. patens* in the second row, which seemed to have led to *S. alterniflora* dominating the patch and perhaps hence leading to A_1 being higher in patch 1 than *S. patens*. The opposite is true of the first row. Overall, changing the initial abundance of the different species has had a great impact on the final equilibrium reached, but many remained with 3 or 4 stable populations with abundances greater than 0.

Parameter/State Variables	Value
g_a	0.3
g_p	0.99
K_a	2.2
K_p	13
d_a	0.7
d_p	0.7
R_a	1
R_p	0.009255
ϕ_1	1
ϕ_2	0.01
P_1	100
P_2	100
A_1	100
A_2	100

Table 1: General Parameter and state variable value for computing solution

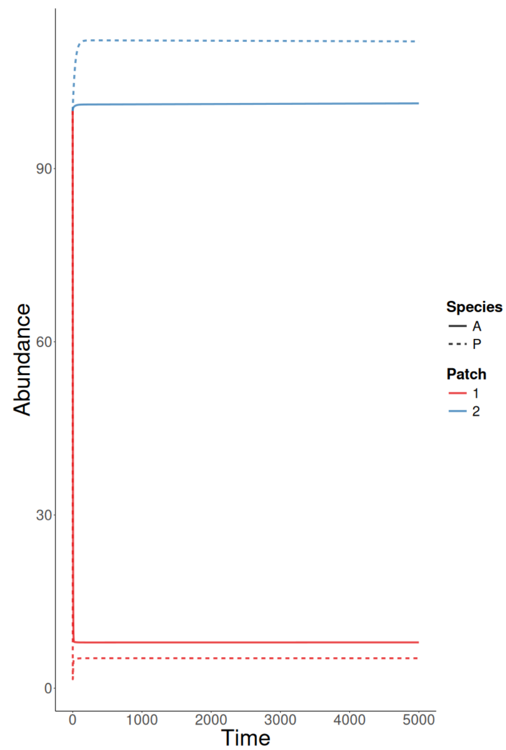

Figure 1: General solutions with the value outline in Table 2

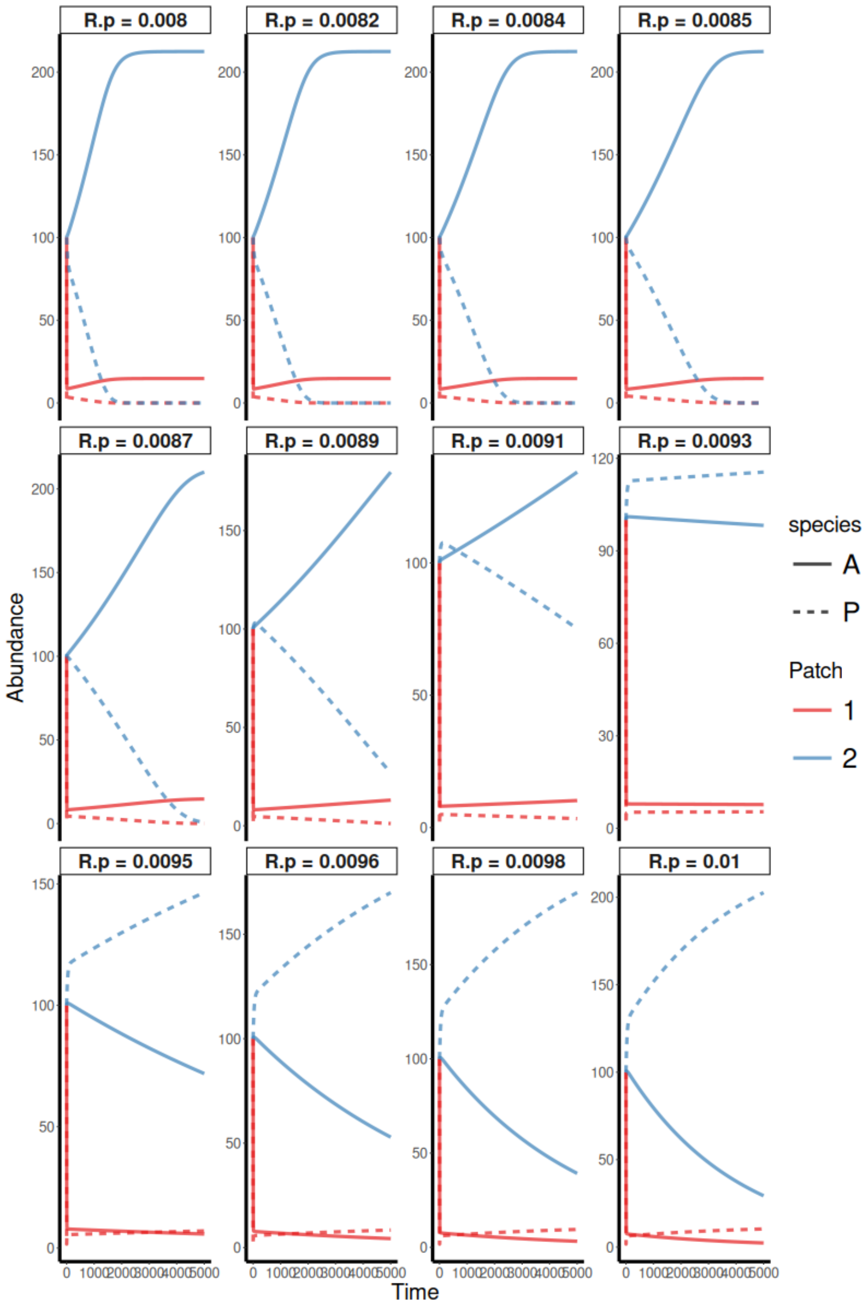

Figure 2: General solution of Figure 1 under different values of R_p

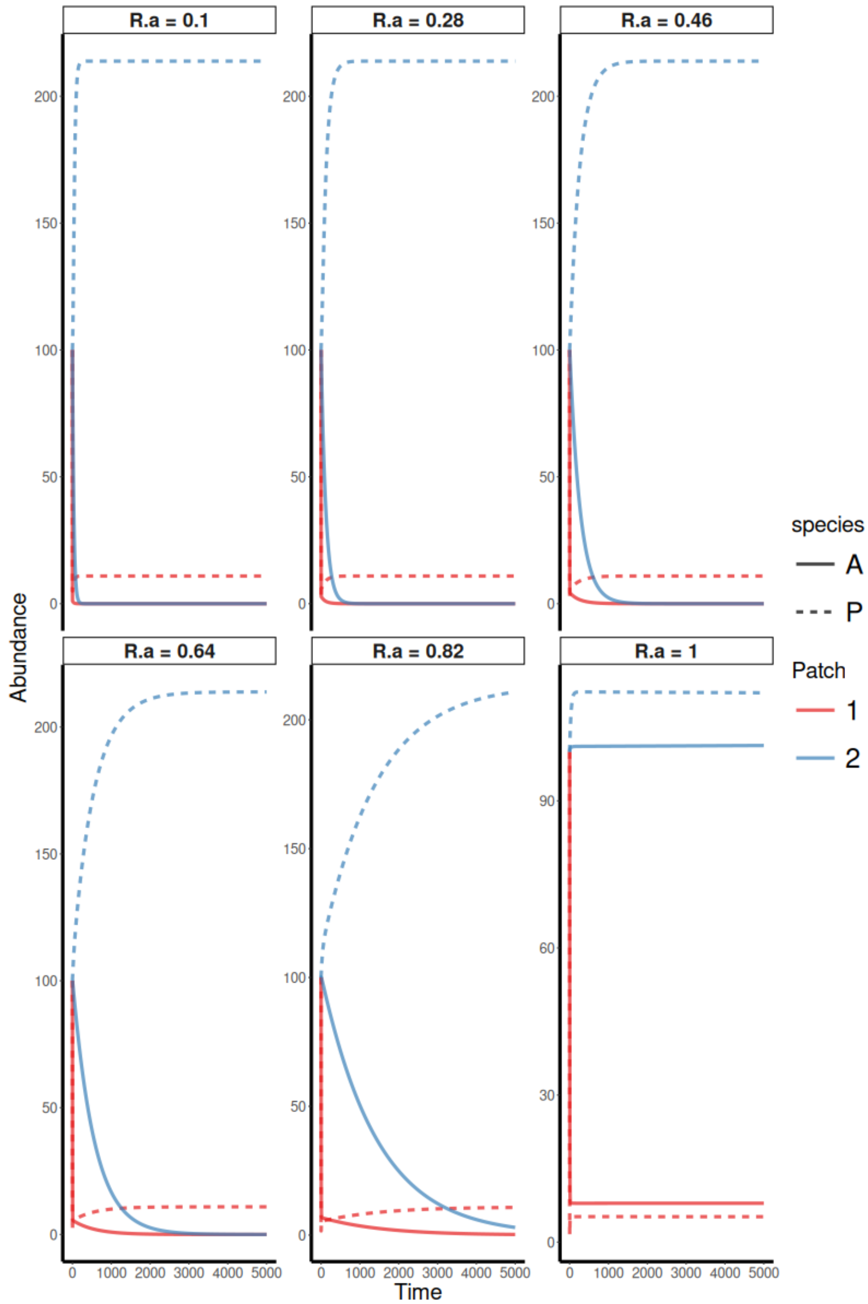

Figure 3: General solution of Figure 1 under different values of R_a

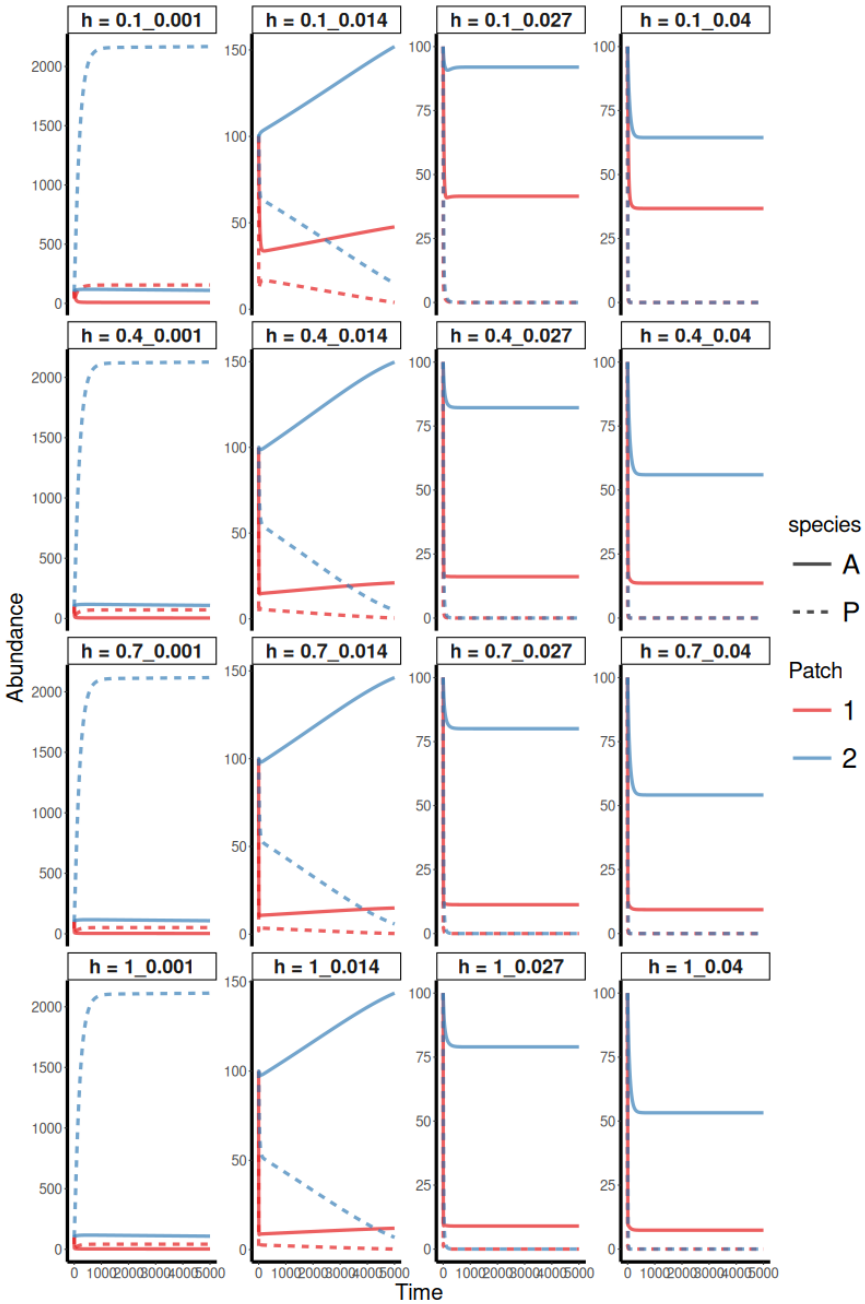

Figure 4: General solution of Figure 1 under different values of h_i. H is a vector of (h_1, h_2)

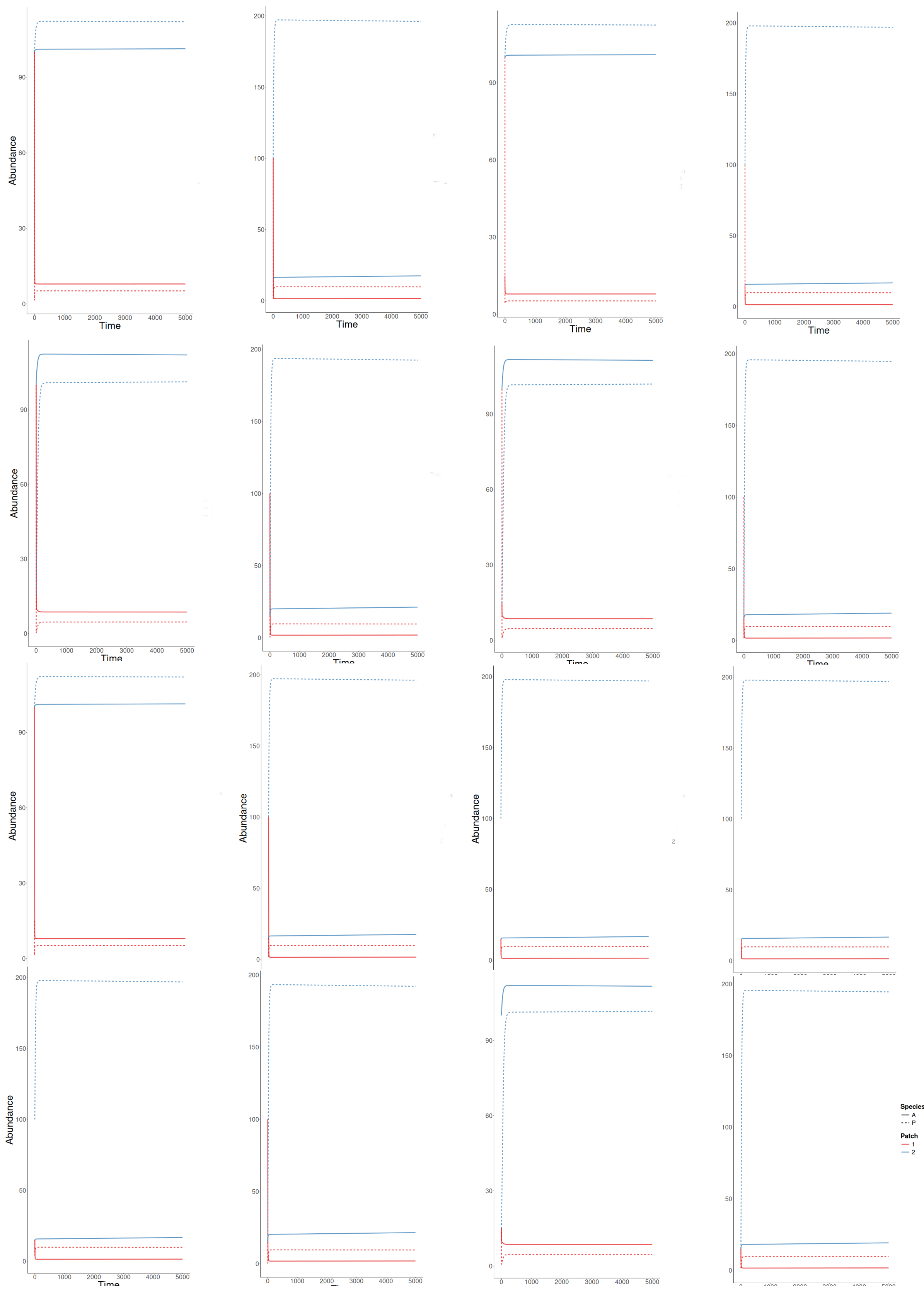

Figure 5: Plot of solution in Figure 1 under different initial conditions. General array of initial condition for these plot follows Table . Of course, it should have been in the graph, but I do not have time for trivial matter.

(100,100,100,100)	(100,100,100,15)	(100,100,15,100)	(100,100,15,15)
(100,15,100,100)	(100,15,100,15)	(100,15,15,100)	(100,15,15,15)
(15,100,100,100)	(15,100,100,15)	(15,100,15,100)	(15,100,15,15)
(15,15,100,100)	(15,15,100,15)	(15,15,15,100)	(15,15,15,15)

Table 2: Initial condition value for the Figure 5, where the vector is (A1,A2,P1,P2

4.2 Ecological Coherent Equilibrium

It was not possible to generate the solution that tended towards a coherent ecological equilibrium with this model, or at least not exactly. The closest solution found to that is the one that was analyzed above. Whether the model simply makes it impossible to find, or the right parameters were not found, is debatable. That being said, many parameter combinations were trialed, manually and automatically, with no success. The particular solution I was searching for was a globally stable full coexistence equilibrium system where these general rules were true:

$$P_2 > A_2 \ , \ A_2 < A_1 \ , \ A_1 > P_1 \approx 0$$

which is true for the solution shown above, but *S. alterniflora* is lower in patch 2 and higher in patch 1. Instead, the solutions for coexistence tend to have *S. alterniflora* dominate both patches, which, at least for the second patch, makes no ecological sense given the presence of *S. patens*. This tells me that the model underestimates competition. Whereas *S. alterniflora* is better suited (in relation to abiotic stress) to grow in high marsh than *S. patens*, it should be outcompeted by *S. patens* despite that advantage. Perhaps the model should include increased mortality of plants due to competition rather than just a decrease in growth. Although I am not certain that this equilibrium is unreachable with the current model format, since the competition is modeled using the relationship between $K_{a,p}$ and $g_{a,p}$ values for different species, I tried to skew competitive outcomes by maximizing K_p and g_p but to no great success. I believe the fact that this model was analyzed as a two-patch model is also part of the problem. If we divide the marsh into two patches, from center to center lies the same distance as if we were comparing the second and fifth patches in a marsh divided into six patches. That being said, in the model, there is less movement between the latter case. The increased colonization of plants from the adjacent patch, I believe, affects the model in a way that makes finding the ecologically coherent solution more difficult. If we decide to go further with this project, I believe finding whether this model can produce this outcome is the primary concern and might push us to reconsider the model. The other possibility is for $P_2 > A_2$ but $P_1 > A_1$, which also does not make any sense.

5 Conclusion

The report outlines the first step in the analysis of a model of *Spartina alterniflora* and *Spartina patens* distribution and interaction across a discretized salt marsh. Although the model could produce solutions resembling ecologically expected results, it was unable to generate a globally stable, ecologically coherent equilibrium based on our analysis. Using a matrix representation of the system, the trivial equilibrium was shown to always be unstable when the growth rate exceeds the death rate, while the non-trivial equilibrium (full coexistence) was demonstrated to exist but exhibited instability when interspecies competition dominated and self-regulation was insufficient to stabilize the system. The existence of this non-trivial equilibrium was also confirmed numerically. It was shown that, under the non-trivial equilibrium case, solutions were impacted as expected by species resistance and patch stress. However, this equilibrium was not globally stable, as changes in initial state variables led to different equilibrium outcomes. The two-patch representation of the model appears to influence the inability to achieve the ecologically expected equilibrium case, though other factors, such as the underestimation of competition effects, might also be contributing.

References

[Bertness, 1991] Bertness, M. D. (1991). Zonation of Spartina Patens and Spartina Alterniflora in New England Salt Marsh. *Ecology*, 72(1):138–148.

[Bertness and Ellison, 1987] Bertness, M. D. and Ellison, A. M. (1987). Determinants of Pattern in a New England Salt Marsh Plant Community. *Ecological Monographs*, 57(2):129–147. _eprint: https://esajournals.onlinelibrary.wiley.com/doi/pdf/10.2307/1942621.

[Caswell, 2001] Caswell, H. (2001). *Matrix Population Models: Construction, Analysis, and Interpretation.* Sinauer Associates, 2nd edition.

[Deimling, 1985] Deimling, K. (1985). *Nonlinear Functional Analysis*, volume 3 of *Springer-Verlag Texts in Applied Mathematics.* Springer-Verlag.

[Hirsch et al., 2013] Hirsch, M. W., Smale, S., and Devaney, R. L. (2013). *Differential Equations, Dynamical Systems, and an Introduction to Chaos.* Academic Press, 3rd edition.

[Horn and Johnson, 2012] Horn, R. A. and Johnson, C. R. (2012). *Matrix Analysis.* Cambridge University Press, 2nd edition.

[Irving A. Mendelssohn and James T. Morris, 2000] Irving A. Mendelssohn and James T. Morris (2000). Eco-physiological controls on the productivity of Spartina alterniflora Loisel. In M. P. Weinstein and Daniel A. Kreeger, editors, *Concepts and Controversies in Tidal Marsh Ecology*, pages 59–80. Springer Science & Business Media. Google-Books-ID: 1fSONFqECloC.

[Karline Soetaert, 2009] Karline Soetaert (2009). *rootSolve: Nonlinear root finding, equilibrium and steady-state analysis of ordinary differential equations.* R package 1.6.

[Karline Soetaert et al., 2010] Karline Soetaert, Thomas Petzoldt, and R. Woodrow Setzer (2010). Solving differential equations in R: Package deSolve. *Journal of Statistical Software*, 33(9):1–25.

[Khalil, 2002] Khalil, H. K. (2002). *Nonlinear Systems.* Prentice Hall, 3rd edition.

[Kreyszig, 1989] Kreyszig, E. (1989). *Introductory Functional Analysis with Applications.* Wiley.

[Levins, 1969] Levins, R. (1969). Some Demographic and Genetic Consequences of Environmental Heterogeneity for Biological Control. *Bulletin of the Entomological Society of America*, 15(3):237–240.

[MacArthur and Wilson, 2001] MacArthur, R. H. and Wilson, E. O. (2001). *The theory of island biogeography.* Princeton University Press, Princeton.

[Mark D. Bertness and Steven C. Pennings, 2000] Mark D. Bertness and Steven C. Pennings (2000). Spatial variation in process and pattern in salt marsh plant communities in eastern North America. In Weinstein, M. P. and Kreeger, D. A., editors, *Concepts and Controversies in Tidal Marsh Ecology*, pages 39–57. Springer Science & Business Media. Google-Books-ID: 1fSONFqECloC.

[Meyer, 2000] Meyer, C. D. (2000). *Matrix Analysis and Applied Linear Algebra.* SIAM.

[Murray, 2002] Murray, J. D. (2002). *Mathematical Biology I: An Introduction*, volume 17 of *Interdisciplinary Applied Mathematics.* Springer, 3rd edition.

[Pickett and White, 1985] Pickett, S. T. A. and White, P. S. (1985). *The Ecology of Natural Disturbance and Patch Dynamics.* Elsevier.

[R Core Team, 2023] R Core Team (2023). R: A Language and Environment for Statistical Computing.

[R Core Team, 2024] R Core Team (2024). *R: A Language and Environment for Statistical Computing.* R Foundation for Statistical Computing, Vienna, Austria.

[Turner, 1989] Turner, M. G. (1989). Landscape Ecology: The Effect of Pattern on Process. *Annual Review of Ecology and Systematics*, 20:171–197. Publisher: Annual Reviews.

[Wiens, 1976] Wiens, J. A. (1976). Population Responses to Patchy Environments. *Annual Review of Ecology and Systematics*, 7:81–120. Publisher: Annual Reviews.